

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and validated **financial risk models** including **derivative valuation models** and **risk measurement models** using **Python** and **R** statistical frameworks, delivering **validation reports** for enterprise banking systems supporting **€50M+ daily transaction volumes**.
- Prepared **mathematical modelling documentation** and **validation reports** for **VaR and behavioural models** across interconnected **ERP and financial risk systems**, collaborating with model developers and stakeholders to track **validation actions to completion** with **100% SLA adherence**.
- Executed **risk analysis and scenario analysis** on production financial systems using **SQL (PostgreSQL, Oracle)** and **Python data stack (pandas, NumPy, scikit-learn)** to validate model assumptions, reducing **model validation cycle time by 30%** through automated testing pipelines.
- Provided **quantitative support** to the Financial Risk team by building **Power BI dashboards** and **Tableau visualizations** for model performance tracking, risk reporting, and stakeholder communication across **derivative pricing, xVA, and risk measurement workflows**.
- Applied **statistical methods, hypothesis testing, and exploratory data analysis** (MSc Data Analytics) to validate **financial product risk models** and behavioural models in a banking environment, leveraging **data warehousing and ETL** techniques to prepare clean datasets for model backtesting.
- Supported change-control management and post-implementation validation of **financial risk software** integrations across banking systems, coordinating **UAT** and presenting **validation findings** to model governance committees with recommendations for model refinement and **regulatory compliance (GDPR, financial industry standards)**.

SKILLS

Quantitative & Financial Risk Modelling – derivative valuation models, VaR models, xVA models, behavioural models, risk measurement models, mathematical modelling, scenario analysis, backtesting.

Programming & Statistical Analysis – Python, R, pandas, NumPy, scikit-learn, hypothesis testing, statistical methods, exploratory data analysis, time-series analysis.

Data & Business Intelligence – Power BI, Tableau, SQL (PostgreSQL, Oracle, MySQL), data modelling, data warehousing, ETL, DAX, Power Query.

Financial Domain & Risk – derivative pricing, risk analysis, financial products, banking industry risk frameworks, SWIFT messaging (MT/MX), ISO 20022, SEPA, payments processing, cross-border payments, financial risk software, regulatory compliance (GDPR).

Enterprise Systems & Integrations – ERP systems, enterprise business systems, REST APIs, system integrations, change-control management, UAT coordination, CI/CD (Jenkins, Git), AWS (EC2, S3, Lambda, CloudWatch).

Collaboration & Documentation – validation report preparation and presentation, stakeholder engagement, cross-functional team collaboration, technical documentation, Jira, Confluence.

Analytical & Professional Qualifications – MSc Data Analytics (NCI, Dublin), BE Computer Science, IELTS Academic Band 7, Python Programming (University of Michigan), Web Development (UC Davis).

EDUCATION AND TRAINING

PROJECTS

Financial Risk Model Validation Framework & Derivative Pricing Model Audit (HCL Technologies)

- Architected an end-to-end **financial risk model validation framework** using **Python, R, and SQL** to validate **derivative valuation models** and **risk measurement models** across a tier-1 banking client's production environment, reducing model validation cycle time from **8 weeks to 5.6 weeks (30% improvement)**.
- Built automated **backtesting pipelines** in **Python (pandas, NumPy, scikit-learn)** to stress-test **VaR and xVA models** against historical market data, generating **statistical validation reports** with hypothesis-testing results, **P&L; attribution analysis**, and model assumption sensitivity matrices.
- Developed **Power BI dashboards** and **Tableau interactive visualizations** for model performance tracking, **risk analysis and scenario analysis results**, and validation status reporting, enabling model governance committees to review **validation findings** and approve **derivative pricing model** updates within **2-week SLA windows**.
- Collaborated with **model developers** and **key stakeholders** (risk managers, business analysts, regulators) to document **validation actions**, track remediation to completion, and prepare formal **validation reports** addressing **behavioural model** calibration, model governance, and **regulatory compliance** requirements (GDPR, financial industry risk standards).
- Integrated validation workflows across **ERP and financial risk systems** using **REST APIs** and **Oracle/PostgreSQL ETL** to synchronize model parameters and test data, ensuring **data integrity and traceability** throughout the validation lifecycle across interconnected banking infrastructure.

Banking Payments Operations & SWIFT/ISO 20022 Risk Monitoring Dashboard (HCL Technologies)

- Designed and maintained a **financial risk software monitoring solution** for a banking client's **SWIFT messaging and payments operations** platform, tracking **VaR exposure** and **derivative pricing impacts** on transaction flows for **€50M+ daily payment volume**, achieving **99.95% system uptime** through **Splunk and Dynatrace monitoring**.
- Applied **mathematical modelling and statistical analysis** to validate **ISO 20022 message flows** and **SEPA/cross-border payment risk models**, ensuring **regulatory compliance** with PSD2 and GDPR requirements, reducing **payment reconciliation errors by 22%** through automated data validation rules.
- Provided **quantitative support** to risk and compliance teams by building **Power BI risk dashboards** and **SQL-based scenario analysis reports** on payment flows, **derivative exposure**, and **behavioural patterns** in customer transactions, enabling rapid decision-making during operational incidents.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)